

Comparison between conductor, semiconductor, and insulator

Parameter	Conductor	Semiconductor	Insulator
Definition	A material that allows electric current to flow through them easily	A material that has conductivity in between conductors and insulators.	Materials that do not allow the electric current to flow through them.
Resistance (R)	Low	Intermediate	Very High
Resistivity (ρ)	Low	Intermediate	Very High
Conductance (G)	High	Intermediate	Very low
Conductivity (σ)	High	Intermediate	Very low
Valance Electrons in Outermost Orbit	Less than 4	4	More than 4
Forbidden Energy Gap (The gap between the valence band and the conduction band is referred to as the forbidden gap.)	No energy gap i.e. the conduction band overlap the valance band.	Small energy gap	Very large energy gap
Examples	Metals like silver, gold, copper, aluminium etc	Silicon, Germanium, Gallium, Arsenide etc	Air, Mica, Glass, Paper, Porcelain, Wood etc
Application	In the manufacturing of conducting wires and cables.	In the manufacturing of solid state electronic devices like ICs, diodes, transistors etc.	Used for providing insulation electrical and electronic devices, for preventing electric shock etc.

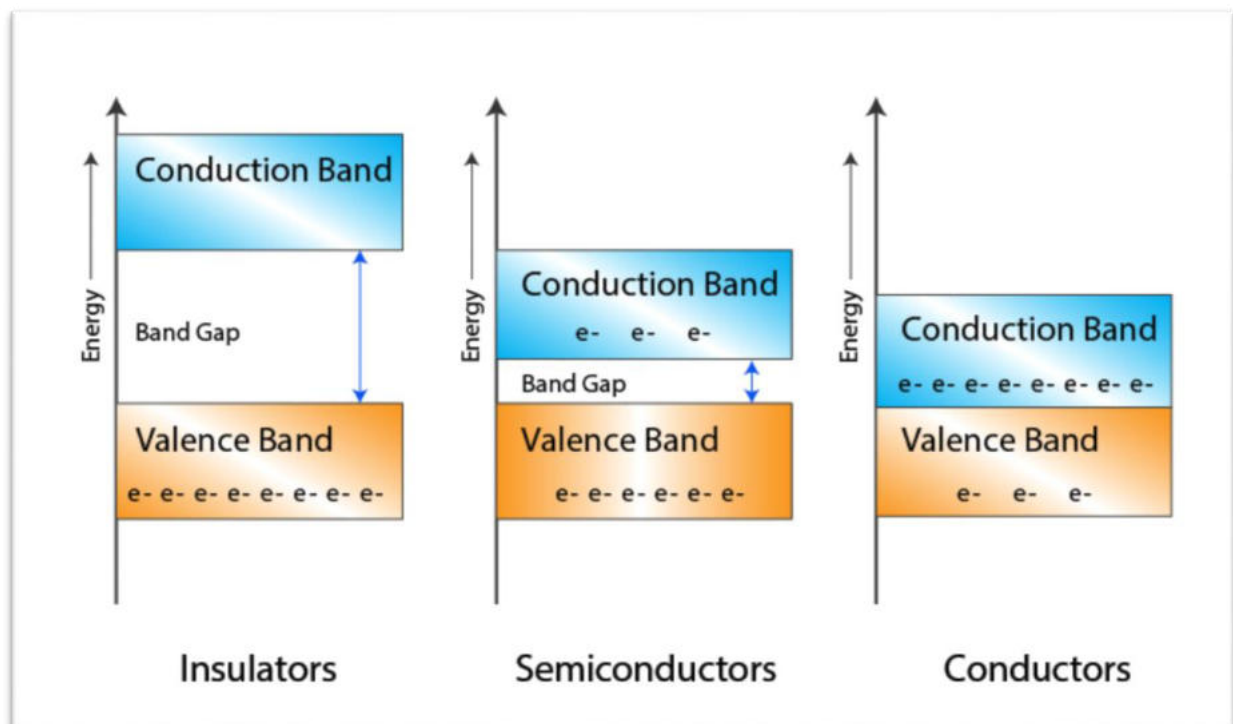
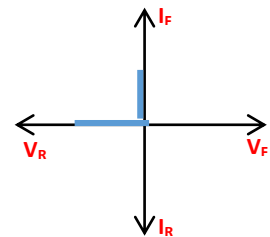
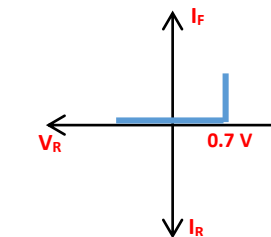
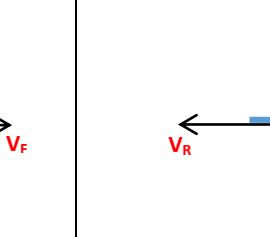


Figure: Band diagram of insulators, semiconductors, and conductors

Comparison Ideal, Second and Third Approximation

Parameter	IDEAL		SECOND		THIRD	
Equivalent Circuits	Forward Biased	Reverse Biased	Forward Biased	Reverse Biased	Forward Biased	Reverse Biased
						
Diode Characteristics						
Bulk Resistance (R_B)	Not Included		Not Included		Included	
Application	Used for Quick analysis OR Trouble Shooting		Used for Analysis at Technical Level		Used for Higher Level Analysis	

Comparison Zener and Avalanche Breakdown

Parameter	Zener Breakdown	Avalanche Breakdown
Breakdown Voltage	5 to 8 Volt	Higher than 8 Volt
Principle	Very High Electric field developed across the junction	Carrier multiplication
VI Characteristics	Very Sharp	Not to Sharp
Effect of Temperature on V_{BR}	V_{BR} decreases with increase in temp.	V_{BR} increases with increase in temp.
Temperature Co-efficient	Negative	Positive
Occurs For	Heavily Doped Diodes	Lightly Doped Diodes